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UC Method:-

The method to find the solution of above type of equations

Step 1: Finding the complementary solution y_c.
Step 2: Finding the particular solution y_p.
Step 3: Taking the derivative of y_p till 1st or 2nd same as the L.H.S derivative is taken.
Step 4: Putting y_p derivates & y_p on L.H.S and make them equal to L.H.S derivative is taken.
Step 5: Finding the values of unknown term & put that term value in y_p.
Step 6: Write the general solution of the given equation by $y = y_c + y_p$.

Example 1: $y'' + 4y' - 2y = 2x^2 - 3x + 6$.

Sol:-

For complementary solution.

let $y = e^{mx}$

$y' = me^{mx}$, $y'' = m^2 e^{mx}$.

Putting y, y', y'' in $y'' + 4y' - 2y = 0$.

we get

$$m^2 e^{mx} + 4me^{mx} - 2e^{mx} = 0.$$

$$e^{mx}(m^2 + 4m - 2) = 0.$$

$$m^2 + 4m - 2 = 0. \quad (\text{Auxillary Eqn})$$

Using quadratic formula

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-4 \pm \sqrt{16 - 4(1)(-2)}}{2(1)}$$

$$m = \frac{-4 \pm 2\sqrt{6}}{2} = \frac{2(-2 \pm \sqrt{6})}{2}$$

$$m = -2 \pm \sqrt{6}.$$

$$m_1 = -2 + \sqrt{6}, \quad m_2 = -2 - \sqrt{6}.$$

So,

$$y_1 = e^{(-2+\sqrt{6})x}, \quad y_2 = e^{-(2+\sqrt{6})x}.$$

$$y_1 = e^{(-2+\sqrt{6})x}, \quad y_2 = e^{-(2+\sqrt{6})x}.$$

$$y_c = C_1 e^{(-2+\sqrt{6})x} + C_2 e^{(-2+\sqrt{6})x}.$$

For Particular Solution

$$y_p = Ax^2 + Bx + C \quad \text{---} (1).$$

$$y_p' = 2Ax + B, \quad y_p'' = 2A.$$

Now

Substituting

$$y'' + 4y' - 2y = 2x^2 - 3x + 6.$$

$$2A + 4(2Ax + B) - 2(Ax^2 + Bx + C) = 2x^2 - 3x + 6$$

$$2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6$$

comparing coefficients

$x^2 \Rightarrow -2A = +2$ $\quad x' \Rightarrow 8A - 2B = -3$

$A = -1$ $\quad 8(-1) - 2(B) = -3$

$-8 - 2B = -3.$

$x^0 \Rightarrow 2A + 4B - 2C = 6$ $\quad -2B = 5$

$2(-1) + 4\left(\frac{-5}{2}\right) - 2C = 6.$ $\quad B = \frac{-5}{2}.$

$-2 - 10 - 2C = 6.$

$-2C = 18.$

$C = -9.$

Putting A, B & C in eq/①.

$$y_p = (-1)x^2 + \left(\frac{-5}{2}\right)x + (-9)$$

$$y_p = -x^2 - \frac{5}{2}x - 9.$$

Hence, the general solution is:

$$y = y_c + y_p = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{-(2+\sqrt{6})x} - x^2 - \frac{5x}{2} - 9.$$

Example 2.

$$y'' - y' + y' = 2\sin 3x.$$

For complementary Solution

Let, $y = e^{mx}$.

$y' = me^{mx}$

$y'' = m^2 e^{mx}$.

putting y, y', y'' in $y'' - y' + y = 0$. So we get

$$m^2 e^{mx} - me^{mx} + e^{mx} = 0.$$

$$e^{mx}(m^2 - m + 1) = 0. \quad , e^{mx} \neq 0.$$

$m^2 - m + 1 = 0$ (Auxilliary Equation).

Using quadratic formula.

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{+1 \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)}.$$

$$m = \frac{1 \pm \sqrt{-3}}{2} = \frac{1 \pm \sqrt{3}i}{2}$$

$$m = \frac{1}{2} \pm \frac{\sqrt{3}i}{2} \implies \alpha = \frac{1}{2}, \beta = \frac{\sqrt{3}}{2}$$

Putting α, β in this eq.

So, $y = e^{\alpha x}\left[C_1 \cos \beta x + C_2 \sin \beta x\right]$.

$$y_c = e^{x/2}\left[C_1 \cos \frac{\sqrt{3}x}{2} + C_2 \sin \frac{\sqrt{3}x}{2}\right]$$

For Particular Solution :-

let,

$$y_p = A \cos 3x + B \sin 3x$$

$$y_p' = -3A \sin 3x + 3B \cos 3x.$$

$$y_p'' = -9A \cos 3x - 9B \sin 3x.$$

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So, putting y_p, y_p', y_p'' in eq (1).

$$-9A\cos 3x - 9B\sin 3x + 3A\sin 3x - 3B\cos 3x + A\cos 3x + B\sin 3x = 2\sin 3x$$

$$-8A\cos 3x + -8B\sin 3x + 3A\sin 3x - 3B\cos 3x = 2\sin 3x$$

(comparing coefficient)

$$\cos 3x(-8A - 3B) + \sin 3x(-8B + 3A) = 2(\sin 3x)$$

comparing coefficients

$$\cos 3x \Rightarrow -8A - 3B = 0 \quad \text{— (A)}$$

$$\sin 3x \Rightarrow -8B + 3A = 2 \quad \text{— (B)}$$

Using equation A

$$-8A = 3B$$

$$A = -\frac{3B}{8}$$ Putting in B

$$A = -3\left(\frac{-16}{8}\right) - 8B - \frac{9B}{8} = 2$$

$$\frac{64B - 9B}{8} = 2$$

$$-73B = 16$$

$$B = -\frac{16}{73}$$

Putting values of B in A.

$$A = -\frac{3(-16/73)}{8}$$

$$A = \frac{6}{73}$$

Putting value of A & B in A

$$y_p = \frac{6}{73}\cos 3x - \frac{16}{73}\sin 3x$$

Hence, the general solution is

$$y = c_1 e^{x/2}\cos\frac{\sqrt{3}}{2}x + c_2 e^{x/2}\sin\frac{\sqrt{3}}{2}x + \frac{6}{73}\cos 3x - \frac{16}{73}\sin 3x.$$

Example 3:

$$y'' - 2y' - 3y = \underbrace{4x - 5}_{\text{Polynomial}} + \underbrace{6xe^{2x}}_{\text{exponential}}$$

Solution:-

$2xe^{2x} + 1e^{2x}$

For complementary solution, So

Let, $y = e^{mx}$.

$y' = me^{mx}$, $y'' = m^2 e^{mx}$.

putting y, y'' & y' in $y'' - 2y' - 3y = 0$, So we get

$$m^2 e^{mx} - 2me^{mx} - 3e^{mx} = 0.$$

$$e^{mx}(m^2 - 2m - 3) = 0.$$

$$m^2 - 2m - 3 = 0. \quad (e^{mx} \neq 0)$$

(Auxillary Solution).

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{4 - 4(1)(-3)}}{2(1)}$$

$$m = \frac{2 \pm 4}{2} = \frac{2+4}{2}, \quad m$$

$$m_1 = 3 \quad , \quad m_2 = -1$$

So,

$$y_1 = e^{3x}, \quad y_2 = e^{-x}$$

$$y_C = c_1 e^{3x} + c_2 e^{-x}.$$

For Particular Solution :-

$$g(x) = g_1(x) + g_2(x) . = \text{polynomial} + \text{exponential}.$$

So,

$$y_P = y_{P_1} + y_{P_2} \quad \text{— (1)}.$$

$$y_{P_1} = Ax + B, \qquad y_{P_2} = Cx e^{2x} + De^{2x}.$$

Putting y_1 & y_2 in eq (1).

$$y_P = Ax + B + Cx e^{2x} + De^{2x} \quad \text{— (2)}$$

$$y_P' = A + C\{2x e^{2x} + e^{2x}\} + 2De^{2x}.$$

$$y_P'' = 0 + 2C\{2x e^{2x} + 2e^{2x} + 2e^{2x}\} + 4De^{2x}.$$

$$y_P'' = 4Cx e^{2x} + 4Ce^{2x} + 4Ce^{2x} + 4De^{2x}$$

$$4Cxe^{2x} + 2Ce^{2x} + 4Ce^{2x} + 4De^{2x} - 2\{A + 2Cxe^{2x} + Ce^{2x} + 2De^{2x}\} - 3\{Ax + B + Cxe^{2x} + De^{2x}\}$$

$$= 4x - 5 + 6xe^{2x}$$

$$= 4Cxe^{2x} + 6Ce^{2x} + 4De^{2x} - 2A - 4Cxe^{2x} - 2Ce^{2x} - 4De^{2x} - 3Ax - 3B - 3Cxe^{2x} - 3De^{2x} = 4x - 5 + 6xe^{2x}$$

$$= -3Ax - 3B - 2A - 3Cxe^{2x} + 2Ce^{2x} - 3De^{2x} = 4x - 5 + 6xe^{2x}.$$

$$= -3Ax - 3B - 2A - 3Cxe^{2x} + 2Ce^{2x} - 3De^{2x} = 4x - 5 + 6xe^{2x}$$

Comparing coefficients

$x \Rightarrow -3A = 4$

$$A = -\frac{4}{3}$$

$x^0 \Rightarrow -3B - 2A = -5$

$$-3B - 2\left(-\frac{4}{3}\right) = -5$$

$$-3B + \frac{8}{3} = -5$$

$$-3B = -\frac{23}{3}$$

$$B = \frac{23}{9}$$

$e^{2x} \Rightarrow 2C - 3D = 0$

$$2(-2) - 3D = 0$$

$$-3D = +4$$

$$D = -\frac{4}{3}$$

$xe^{2x} \Rightarrow -3C = 6$

$$C = -2$$

Putting values of A, B, C, D in eq 2

$$y_p = -\frac{4}{3}x + \frac{23}{9} - 2xe^{2x} - \frac{4}{3}e^{2x}.$$

Hence, the general solution

$$y = C_1e^{3x} + C_2e^{-x} - \frac{4}{3}x + \frac{23}{9} - 2xe^{2x} - \frac{4}{3}e^{2x}.$$

Example #4: Find a particular solution

$$y'' - 5y' + 4y = 8e^x \quad -A)$$

For particular solution

Let $y_p = Axe^x \rightarrow 1)$

$$y_p' = A\{xe^x + e^x\} = Axe^x + Ae^x$$

$$y_p'' = Axe^x + Ae^x + Ae^x = Axe^x + 2Ae^x$$

Now putting y, y_p' and y_p'' in eq A.

$$Axe^x + 2Ae^x - 5Axe^x - 5Ae^x + 4Axe^x = 8e^x$$

$$Axe^x - Axe^x - 3Ae^x = 8e^x$$

$$-3Ae^x = 8e^x$$

Comparing coefficients.

$$e^x = -3A = 8$$

$$A = \frac{-8}{3}$$

Example #5: Find a particular solution of

a) $y'' - 8y' + 25y = 5x^3 e^{-x} - 7e^{-x}$

For complementary solution.

$m^2 - 8m + 25 = 0$

$m = \frac{8 \pm \sqrt{(8)^2 - 4(1)(25)}}{2(1)}$

$m = \frac{8 \pm \sqrt{-36}}{2} \Rightarrow m = \frac{8 \pm 6i}{2}$.

$m = 4 \pm 3i \Rightarrow \alpha = 4, \beta = 3$.

So,

$y_c = e^{4x}[c_1 \cos 3x + c_2 \sin 3x]$

For particular Solution.

$g(x) = (5x^3 - 7)e^{-x}$

So,

$y_p = (Ax^3 + Bx^2 + Cx + D)e^{-x}$.

b) $y'' + 4y = x \cos x$.

For complementary Solution.

$m^2 + 4 = 0 \Rightarrow m^2 = -4$

$m = \pm 2i \Rightarrow \alpha = 0, \beta = 2i$

So, $y_c = e^0[c_1 \cos 2x + c_2 \sin 2x]$

$y_c = c_1 \cos 2x + c_2 \sin 2x$.

For particular solution.

Let $y_p = (Ax + B)\cos x + (Cx + D)\sin x$

Example #6:- Determine the form of particular Solution.

$$y'' - 9y' + 14y = 3x^2 - 5\sin 2x + 7xe^{6x}.$$

y_p for $3x^2$: $y_{p_1} = Ax^2 + Bx + C$

y_p for $-5\sin 2x$: $y_{p_2} = D\cos 2x + E\sin 2x$

y_p for $7xe^{6x}$: $y_{p_3} = (Fx + G)e^{6x}$.

Hence,

$$y_p = y_{p_1} + y_{p_2} + y_{p_3} = Ax^2 + Bx + C + D\cos 2x + E\sin 2x + (Fx+G)e^{6x}$$

For complementary Solution

$$m^2 - 9m + 14 = 0$$

$$m^2 - 7m - 2m + 14 = 0$$

$$m(m-7) - 2(m-7) = 0$$

$$(m-2)(m-7) = 0$$

$$m = 2 \mid m = 7.$$

So $y_c = c_1 e^{2x} + c_2 e^{7x}$.

Example #7: $y'' - 2y' + y = e^x$ -A)

For complementary Solution.

let $y = e^{mx}$

$y' = me^{mx}$, $y'' = m^2 e^{mx}$

putting y, y', y'' in $y'' - 2y' + y = 0$

we get : $m^2 e^{mx} - 2me^{mx} + e^{mx} = 0$

$$e^{mx}(m^2 - 2m + 1) = 0 \qquad e^{mx} \neq 0$$

$m^2 - 2m + 1 = 0$ (Auxiliary equation).

$m^2 - m - m + 1 = 0$

$m(m-1) - 1(m-1) = 0$

$(m-1)(m-1) = 0$

$m_1 = m_2 = 1.$

So, $y_c = c_1 e^x + c_2 x e^x$

For particular Solution.

let $y_p = Ax^2 e^x$

$y_p' = 2Axe^x + Ax^2 e^x$

$y_p'' = 2Ae^x + 2Axe^x + 2Axe^x + Ax^2 e^x$

$y_p''' = 2Ae^x + 4Axe^x + Ax^2 e^x$

Putting y_p, y_p', y_p'' in eq A)-

$2Ae^x + 4Axe^x + Ax^2 e^x - 4Axe^x - 2Ax^2 e^x + Ax^2 e^x = e^x$

$2Ae^x + 2Ax^2 e^x - 2Ax^2 e^x = e^x.$

$2Ae^x = e^x$

comparing coefficients $2A = 1.$

$A = 1/2$

So, $y_p = \frac{1}{2} x^2 e^x.$

Example #8:. Solve the IVP

$y'' + y = 4x + 6\sin x \quad ; y(\pi) = 0, \; y'(\pi) = 2$

For complementary Solution.

$m^2 + 1 = 0$	So, $\alpha = 0, \beta = 1$
$m^2 = -1.$	$y_c = c_1 \cos x + c_2 \sin x$
$m = \pm 1i$	

For particular solution.

Let $y_p = Ax + B + Cx\cos x + Dx\sin x$ — 1)

$y_p' = A + 0 + C\{-x\sin x + \cos x\} + D\{x\cos x + \sin x\}$

$y_p' = A - Cx\sin x + C\cos x + Dx\cos x + D\sin x$

$y_p'' = -C\{x\cos x + \sin x\} - C\{\sin x + B\}\{-x\sin x + B\cos x\}$

$+ D\cos x$

$y_p'' = -Cx\cos x - C\sin x - C\sin x - Dx\sin x + D\cos x + D\cos x$

$y_p'' = -Cx\cos x - Dx\sin x - 2C\sin x + 2D\cos x$

Putting y_p and y_p'' in eq (A)

$-Cx\cos x - Dx\sin x - 2C\sin x + 2D\cos x + Ax + B + Cx\cos x + Dx\sin x$

$= 4x + 10\sin x$

$-2C\sin x + 2D\cos x + Ax + B = 4x + 10\sin x.$

Comparing coefficients.

$-2C = 10$	$A = 4$	$2D = 0$	$B = 0$
$C = -5$		$D = 0$	

Putting A, B, C, D in eq (1)

$y_p = 4x + 0 - 5x\cos x + 0$

$y_p = 4x - 5x\cos x$

Hence, the general solution is

$y = C_1\cos x + C_2\sin x + 4x - 5x\cos x$ → 2)

Using $y = 0$, $x = \pi$

$0 = C_1\cos(\pi) + C_2\sin(\pi) + 4(\pi) - 5(\pi)\cos(\pi)$

Now putting $y=2$ and $x=\pi$.

$$y'(\pi) = -9\pi\sin\pi + C_2\cos\pi + 4 + 5\pi\sin\pi - 5\cos\pi = 2$$

$$\Rightarrow 0 - C_2 + 4 + 0 + 5 = 2$$

$$C_2 = 7$$

So the eq (2) becomes.

$$y = 9\pi\cos x + 7\sin x + 4x - 5x\cos x.$$

Solve :- $y'' - 6y' + 9y = 6x$

Cauchy Euler Equation

↳ Also called equidimensional equation.

→ Any linear DE of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = g(x).$$

OR -

$$x^2 y'' + xy' + y = 0$$

The power of x should be equal to order of y with it e.g $x^{②}y^{②}$ → both are same.

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→ For 2nd order homogenous equation.

$$ax^2\frac{d^2y}{dx^2} + bx\frac{dy}{dx} + cy = 0$$

→ For 2nd order non-homogenous D.E:-

$$ax^2\frac{d^2y}{dx^2} + bx\frac{dy}{dx} + cy = g(x)$$

Method of Solution. If an equation is cauchy euler with variable coeffficients:

Step1: let $y = x^m$.

Step 2: Taking 1st and 2nd derivative of $y = x^m$

$$\frac{dy}{dx} = mx^{m-1}, \quad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

step 3: putting $\frac{d^2y}{dx^2}$, $\frac{dy}{dx}$ and y in general solution.

$$ax^2 \cdot m(m-1)x^{m-2} + bx \cdot mx^{m-1} + cx^m = 0$$

$$am(m-1)x^m + bmx^m + cx^m = 0$$

$$x^m\left(am(m-1) + bm + c\right) = 0, \quad x^m \neq 0$$

So, $am(m-1) + bm + c = 0$

$$am^2 - am + bm + c = 0$$

$$am^2 + (b-a)m + c = 0 \rightarrow \text{cauchy euler equation}$$

Case I: Distinct Real Roots:

→ If m_1 and m_2 are the real roots and $m_1 \neq m_2$.

→ So, $y_1 = x^{m_1}$ and $y_2 = x^{m_2}$

$$y = C_1 x^{m_1} + C_2 x^{m_2}.$$

Example #1:- $x^2 \frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = 0$ — A)

$$y = x^m.$$

$$y' = mx^{m-1}, \quad y'' = m(m-1)x^{m-2}$$

So putting y, y' and y'' in eq (A) we get:-

$$x^2 \cdot m(m-1)x^{m-2} - 2x \cdot mx^{m-1} - 4x^m = 0$$

$$m(m-1)x^m - 2mx^m - 4x^m = 0$$

$$x^m \{m^2 - m - 2m - 4\} = 0 \qquad \because x^m \neq 0$$

So, $m^2 - 3m - 4 = 0$ (cauchy Eulea)

By Quadratic formula.

$$m = \frac{3 \pm \sqrt{9 - 4(1)(-4)}}{2} = \frac{3 \pm 5}{2}$$

$$m_1 = \frac{3+5}{2}, \quad m_2 = \frac{3-5}{2}.$$

$$m_1 = 4, \quad m_2 = -1$$

$$y = C_1 x^4 + C_2 x^{-1}$$

Case II: Repeated Real Roots.

If the roots m_1 and m_2 are equal.

$m_1 = m_2$, when $b^2 - 4ac = 0$

$\Rightarrow y_1 = x^{m_1}$

$\Rightarrow$ For y_2 using $am^2 + (b-a)m + c = 0$

using quadratic formula.

$$m_1 = \frac{-b \pm 0}{2a} = \frac{-b}{2a} \qquad \because b^2 - 4ac = 0$$

As $b = b - a$

$$\text{So, } m_1 = \frac{-(b-a)}{2a}$$

For $p(x)$ = Make the general solution to

$$y'' + \frac{b}{ax} y' + \frac{c}{ax^2} y = 0$$

$$P(x) = \frac{b}{ax}$$

$$y_2 = y_1(x) \int \frac{e^{-\int P x \, dx}}{y_1^2(x)} dx$$

$$y_2 = x^{m_1} \int \frac{e^{-\frac{b}{a}\int \frac{1}{x} dx}}{x^{2m_1}}$$

$$y_2 = x^{m_1} \int \frac{e^{-\frac{b}{a} \ln|x|}}{x^{2m_1}} dx$$

$$y_2 = x^{m_1} \int x^{-\frac{b}{a}} \cdot x^{-2m_1} dx.$$

$$y_2 = x^{m_1} \int x^{-\frac{b}{a}} \cdot x^{-\frac{b}{a}} \cdot x^{1} dx$$

$$y_2 = x^{m_1} \int \frac{1}{x} dx$$

$$y_2 = x^{m_1} \ln x.$$

The general solution is

$$y = c_1 x^{m_1} + c_2 x^{m_1} \ln x.$$

Example #2: $4x^2\frac{d^2y}{dx^2} + 8x\frac{dy}{dx} + y = 0 \rightarrow A)$

Let $y = m^x$.

$$\frac{dy}{dx} = mx^{m-1}, \quad \frac{d^2y}{dx^2} = m(m-1)x^{m-2}$$

Putting y, y', y'' in eq (A).

$$4x^2m(m-1)x^{m-2} + 8xmx^{m-1} + x^m = 0$$

$$4m(m-1)x^m + 8mx^m + x^m = 0$$

$$x^m(4m(m-1) + 8m + 1) = 0$$

$$4m^2 - 4m + 8m + 1 = 0 \qquad \because x^m \neq 0$$

$$4m^2 + 4m + 1 = 0 \quad \text{(cauchy euler equation)}$$

$$(2m+1)^2 = 0$$

$$2m + 1 = 0$$

$$m = -1/2$$

$$y_1 = x^{-1/2}, \quad y_2 = x^{-1/2}\ln x$$

So, $y = c_1x^{-1/2}$ and $c_2x^{-1/2}\ln x$

Case III : Conjugate Complex Roots.

If roots are complex and $b^2 - 4ac < 0$ then:

$$m_1 = \alpha + i\beta, \quad m_2 = \alpha - i\beta$$

As $y = x^m$

$$y_1 = x^{\alpha+i\beta}, \quad y_2 = x^{\alpha-i\beta}$$

$$y = c_1x^{\alpha+i\beta} + c_2x^{\alpha-i\beta}$$

$x^{i\beta} = (e^{\ln x})^{i\beta} = e^{i\beta \ln x}$

Using euler's formula:-

$e^{i\theta} = \cos\theta + i\sin\theta$. $\quad \theta = \beta \ln x$

So, $x^{i\beta} = \cos(\beta \ln x) \pm i\sin(\beta \ln x)$

$y = x^{\alpha}[c_1 x^{i\beta} + c_2 x^{-i\beta}]$

$y = x^{\alpha}[c_1[\cos\beta \ln x + i\sin\beta \ln x] + c_2[\cos\beta \ln x - i\sin\beta \ln x]$

$y = x^{\alpha}[(c_1+c_2)\cos\beta \ln x + i(c_1-c_2)\sin\beta \ln x]$

$y = x^{\alpha}[A\cos\beta \ln x + B\sin\beta \ln x]$.

Example #3: $x^2y'' + 3xy' + 3y = 0$; $y(1)=1$, $y'(1)=-5$

↳ A)

Solution:-

$y = x^m$

$\frac{dy}{dx} = mx^{m-1}$, $\frac{d^2y}{dx^2} = m(m-1)x^{(m-2)}$

Putting in A.

we get : $x^2 m(m-1)x^{m-2} + 3x\,mx^{m-1} + 3x^m = 0$

$m(m-1)x^m + 3mx^m + 3x^m = 0$

$x^m\{m^2 - m + 3m + 3\} = 0$

$m^2 + 2m + 3 = 0 \quad \because x^m \neq 0$

Use quadratic formula

$m = \frac{-2 \pm \sqrt{4 - 4(1)(3)}}{2(1)} = \frac{-2 \pm \sqrt{8}}{2}$

$m = -1 \pm \sqrt{2}\,i$

$\alpha = -1$, $\beta = \sqrt{2}$

So, $y = x^{-1}[c_1 \cos(\sqrt{2}\ln x) + c_2 \sin(\sqrt{2}\ln x)] \quad -1)$

Now putting $y=1,\ x=1$.

$1 = (1)^{-1}[c_1 \cos\sqrt{2}\ln(1) + c_2 \sin\sqrt{2}\ln(1)]$

$1 = c_1 \cos(0) + c_2 \sin(0)$

$c_1 = 1$.

$y' = x^{-1}\{\sqrt{2}(\cos(\sqrt{2}\ln x)) - \sin(\sqrt{2})\ln x\} - (x^{-2})\{\cos(\sqrt{2}\ln x) + c_2 \sin\sqrt{2}\ln x$

$y' = -5,\ x = 1$.

$-5 = (1)^{-1}\{\sqrt{2}(c_2\cos(0) - \sin(0)\} - (1)^{-2}\{\cos(0) + c_2(0)\}$

$-5 = \sqrt{2}c_2 - 1$

$-4 = \sqrt{2}c_2$.

$c_2 = \frac{-4}{\sqrt{2}}$

$c_2 = -2\sqrt{2}$.

So equation (1) becomes:

$y = x^{-1}[\cos(\sqrt{2}\ln x) - 2\sqrt{2}\{\sin(\sqrt{2}\ln x)]$

Cauchy Euler Equation (Non-homogenous).

→ The coefficients ~~or~~ in these examples are variable.

→ Use the transformation $x = e^t$ to make them constants.

→ By using $y = x^m$ we can only find the complementary solution.

Example #6:-

Solve $x^2y'' - xy' + y = \ln x$.

↳ A)

Using the substitution $x = e^t$ or $t = \ln x$.

So $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{1}{x}\frac{dy}{dt}$ — 1)

$\frac{d^2y}{dx^2} = \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dt}\right) + \frac{dy}{dt}\left(\frac{-1}{x^2}\right)$ $\quad \because \frac{d}{dx} = \frac{d}{dt} \times \frac{dt}{dx}$

$\frac{d^2y}{dx^2} = \frac{1}{x}\left\{\frac{d}{dt}\cdot\frac{dt}{dx}\cdot\frac{dy}{dt}\right\} + \frac{dy}{dt}\left(-\frac{1}{x^2}\right)$

$\frac{d^2y}{dx^2} = \frac{1}{x}\left\{\frac{d^2y}{dx^2}\cdot\frac{1}{x}\right\} + \frac{dy}{dt}\left(-\frac{1}{x^2}\right)$ $\quad \because \frac{dt}{dx} = \frac{1}{x}$

$\frac{d^2y}{dx^2} = \frac{1}{x^2}\left(\frac{d^2y}{dx^2} - \frac{dy}{dt}\right)$ → 2)

Now putting (1), (2) and (3) in eq (A).

$\frac{d^2y}{dx^2} - \frac{dy}{dt} - \frac{dy}{dt} + y = t$

$\frac{d^2y}{dx^2} - 2\frac{dy}{dt} + y = t$ → (4)

let $y = e^{mt}$ $\Rightarrow y' = me^{mt}$, $y'' = m^2e^{mt}$

So putting y, y', y'' in equation (4).

$m^2e^{mt} - 2me^{mt} + e^{mt} = 0$

$e^{mt}(m^2 - 2m + 1) = 0$ $\quad \because e^{mt} \neq 0$

$m^2 - 2m + 1 = 0$

Using factorization method.

$m^2 - m - m + 1 = 0$

$m(m-1) - 1(m-1) = 0$

$(m-1)^2 = 0 \Rightarrow m - 1 = 0 \Rightarrow m_1 = 1.$

So, $y_1 = e^t$, $y_2 = te^t$

$y_c = c_1e^t + c_2te^t.$

Now, For particular Solution.

Let $y_p = At + B$.

$y_p' = A$, $y_p'' = 0$

So, eq (4) $\Rightarrow$ $0 - 2A + At + B = t$

$(-2A + B) + At = t$.

comparing coeffiecients

$t^1 \Rightarrow \boxed{A = 1}$ $\qquad t^0 \Rightarrow -2A + B = 0$

$-2(1) + B = 0$

$B = 2$.

So, $y_p = t + 2$

$y = y_c + y_p = c_1 e^t + c_2 t e^t + 2 + t$

Putting $t = \ln x$, $e^t = x$

Hence,

$y = c_1 x + c_2 x \ln x + 2 + \ln x$